
Evaluating Foundation Model for Tabular Data in Industrial Manufacturing

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Abstract

Resistance spot welding (RSW) is one of the most used joining processes in automotive manufacturing. Modern car bodies containing thousands of welded points. Its performance directly impacts manufacturing cost, product quality, and structural integrity. Accurate prediction of weld quality is challenging, as purely physics-based formulas often fail to capture the complexity of real production data, requiring extensive physical testing for validation. Artificial intelligence (AI) offers a data-driven alternative to model these complex interactions, potentially reducing testing effort and improving prediction accuracy.

This work focuses on evaluating the *Tabular Prior-Data Fitted Network* (TabPFN), a foundation model for tabular data prediction. While foundation models have achieved remarkable success in natural language processing and computer vision, equivalent general-purpose models for tabular data remain rare. Assessing their capabilities is therefore of high interest for industrial analytics. As a representative use case, RSW process data is chosen due to its industrial relevance, rich combination of numerical and categorical features, and the importance of accurate weld quality prediction for manufacturing efficiency.

TabPFN’s performance is benchmarked against tuned XGBoost, tuned Random Forest, and a physics-informed XGBoost model that integrates domain specific welding knowledge. The comparison focuses on predictive accuracy and the trade-offs between purely data driven and hybrid physics machine learning approaches. Results show that foundation models can match or exceed established machine learning methods in manufacturing applications, while physics integration remains valuable for interpretability and domain trust.

1. Introduction

Foundation models have achieved state of the art performance in domains such as natural language processing and computer vision by pretraining on large, diverse datasets and applying the resulting model to new tasks without dataset specific retraining. In contrast, few such models exist for tabular data, despite its prevalence in scientific and industrial applications. Most current approaches e.g., gradient boosted decision trees require dataset specific training and hyperparameter tuning.

The Tabular Prior-Data Fitted Network (TabPFN) (Hollmann et al., 2025) is a recent attempt to close this gap. It is pretrained on millions of synthetically generated datasets sampled from a structural causal model prior, covering a wide variety of feature target relationships, noise levels, and data irregularities. At inference time, TabPFN performs both training and prediction in a single forward pass, offering a general purpose alternative to conventional methods for small to medium sized tabular datasets.

TabPFN is evaluated in the context of resistance spot welding (RSW), a representative manufacturing process where weld quality prediction is both economically and technically important. Modern automotive bodies contain several thousand weld points, and the quality of each weld affects structural integrity, crash safety, and production efficiency. Accurate prediction of weld strength is challenging: physical formulas for RSW, while interpretable, often fail to capture the variability present in real production data, where numerous interacting factors, such as material properties, surface conditions, geometric tolerances, and machine characteristics are difficult to measure or model explicitly.

Artificial intelligence (AI) offers a complementary approach by learning predictive models directly from process data, enabling the capture of complex, nonlinear dependencies that are difficult or impossible to model analytically. As Murphy observes, the goal of machine learning is to “develop methods that can automatically detect patterns in data, and then to use the uncovered patterns to predict future data or other outcomes of interest” (Murphy, 2012). This pattern-discovery capability is well aligned with the challenges of

modeling RSW quality, where historical production data can be leveraged to reduce physical testing requirements.

In addition to purely data-driven methods, hybrid approaches that integrate physics-based constraints (Karniadakis et al., 2021) aim to combine predictive accuracy with physical consistency, which is often critical in industrial deployment. This work compares TabPFN to strong tree-based baselines and a physics-guided residual learning approach, providing an empirical assessment of foundation models for tabular data in a realistic manufacturing setting.

1.1. Problem Statement and Contributions

While foundation models have shown strong performance in vision and language, their applicability to tabular data, especially in industrial manufacturing, remains underexplored. In particular, no prior work has systematically benchmarked a tabular foundation model against both conventional machine learning baselines and physics-guided approaches on a realistic manufacturing task.

This study addresses that gap through the following contributions:

- **Benchmarking TabPFN in manufacturing:** Evaluating the Tabular Prior-Data Fitted Network (TabPFN) on real resistance spot welding (RSW) process data, representing a complex, small-to-medium scale industrial dataset.
- **Comparative analysis:** Comparing TabPFN to tuned gradient-boosted decision trees (XGBoost), random forests, and a physics-guided XGBoost residual learning model that incorporates domain-specific welding knowledge.
- **Empirical insights:** Analyzing the performance trade-offs between general purpose foundation models and physics-integrated approaches, providing guidance for model selection in manufacturing quality prediction.

1.2. Paper Organization.

The remainder of this paper is structured as follows: Section 2 reviews related work, including the TabPFN architecture, fundamentals of resistance spot welding, and approaches for integrating physics into machine learning. Section 3 describes the dataset, preprocessing steps, model configurations, and experimental setup. Section 4 presents and discusses the results, focusing on performance comparisons and industrial implications. Finally, Section 6 concludes the paper and outlines directions for future work.

2. Related Work

2.1. TabPFN

The difference between tabular data and data such as images or text is the variety. While in text, where the meaning of a word is consistent across documents, the same value in a table can represent can mean different things.

For this reason, deep learning methods have traditionally struggled with tabular data, making non-deep learning methods, such as tree-based models, the preferred choice.

The Tabular Prior-Data Fitted Network (TabPFN) (Hollmann et al., 2025) takes a different approach. It is designed as a foundation model for tabular prediction that avoids dataset specific training altogether. At its core is the idea of *in-context learning* (ICL), a mechanism first popularized in large language models, where the system learns a prediction algorithm by observing examples rather than by updating its weights for each new dataset (Vaswani et al., 2017).

In TabPFN’s case, this learning ability is acquired through pretraining on millions of synthetic datasets, each generated from a structural causal model (SCM) prior. The prior is constructed to represent a wide range of potential feature-target relationships, noise patterns, and data irregularities, such as missing entries or mixed feature types. By seeing these variations during training, the model develops strategies that can be applied to new datasets without additional finetuning. With this approach, it was created around 100 million synthetic datasets per model training (Hollmann et al., 2025).

TabPFN builds on a transformer architecture adapted for tabular data. Unlike standard transformers, which process input as a single sequence, it explicitly leverages the table structure by assigning each cell its own representation. A two-way attention mechanism allows each cell to attend to other features in the same sample and to all samples of the same feature. This design is invariant to the order of rows and columns and enables efficient generalization to tables larger than those seen during training. In line with its in-context learning setup, TabPFN processes training and test data jointly, performing training and prediction in a single forward pass.

Evaluation on small tabular datasets show, that TabPFN consistently outperformed strong baselines, including both default and tuned configurations of tree-based models such as CatBoost, XGBoost, and Random Forests. The model achieved competitive results across both classification and regression tasks without the need for dataset-specific hyperparameter tuning. Beyond predictive performance, TabPFN also demonstrated efficient advantages, delivering accurate results within seconds on a standard CPU.

Despite its strengths, TabPFN is currently best suited for

datasets of limited size, typically up to around ten thousand samples. Its performance on much larger datasets has not yet been fully explored.

2.2. Resistance Spot Welding

Resistance spot welding (RSW) is considered the dominant joining method for automotive body construction due to its low cost and ease of use. Depending on the vehicle type, a car body is joined with 3,000 to 5,000 weld spots, applied automatically by industrial robots (Pischinger & Seiffert, 2021, p. 179). The principles of RSW have been often described in textbooks and technical literature. In the following, the main aspects relevant to this work are summarized to provide a self-contained description. Later in this section, particular attention is given to the physics-based relationship used in the experiments, which are taken from a specific study on weld strength prediction.

2.2.1. PHYSICAL FUNDAMENTALS

The physical principle underlying RSW is Joule's law. The amount of heat Q required to melt the material is generated directly at the weld location between the joining partners by an electric current I flowing through the electrical resistance R over the welding time t . This relationship is expressed as (Kusch et al., 2022, p. 332):

$$Q = I^2 R t \quad (1)$$

The total resistance R consists of both contact resistances and material resistances. The total resistance R can be

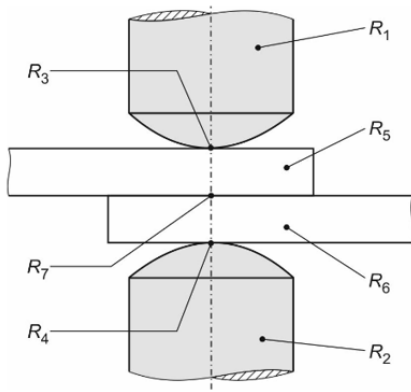


Figure 1. Schematic representation of resistance components in resistance spot welding (Kusch et al., 2022, p. 341).

expressed as:

$$R = R_1 + R_2 + R_3 + R_4 + R_5 + R_6 + R_7 \quad (2)$$

where:

- R_1, R_2 : material resistance of the electrodes,

- R_3, R_4 : contact resistance between electrode and sheet,
- R_5, R_6 : material resistance of the sheet,
- R_7 : contact resistance between the sheets.

- **Contact resistances** R_c are influenced by contaminants and oxide layers on the surfaces, gap distances, component coatings, and sheet lubrication. These resistances decrease as the temperature rises and approach zero once the melting temperature is reached.
- **Material resistances** R_m depend on the current-carrying cross-sectional area, the electrical conductivity, and the temperature of the materials. The material resistance increases with rising temperature until the melting temperature is reached. Once this point is exceeded, a lens-shaped weld nugget begins to form. This increases the conductive cross-sectional area, which in turn causes the material resistance to decrease (Pischinger & Seiffert, 2021, p. 341).

2.2.2. PROCESS DESCRIPTION

The sheets to be welded are arranged in an overlapping configuration, and the electrical current is supplied through copper electrodes that are applied on both sides of the joint with a defined electrode force (Kampker et al., 2018, p. 260). Typically, two to three steel sheets are welded together, with individual sheet thicknesses ranging from 0.5 mm to 3 mm.

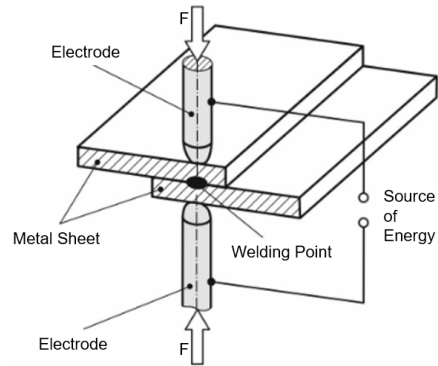


Figure 2. Schematic representation of resistance components in resistance spot welding (Kusch et al., 2022, p. 340).

The welding process can be divided into three phases: closing, welding, and cooling (Schuler & Twrdek, 2019, p. 105).

- **Phase 1 - Closing:** The electrodes are brought together with a predefined force. To ensure that the full electrode force is achieved and the electrodes are fully closed, a squeeze time is applied before the current flow begins.
- **Phase 2 - Welding:** The welding time and current then cause the material to melt, forming a weld nugget.
- **Phase 3 - Cooling:** The electrode force remains active, but no welding current is applied. During the subsequent hold time, the weld nugget is cooled until sufficient strength is reached. After cooling, the electrodes are opened (Schuler & Twrdek, 2019, p. 105).

2.2.3. MECHANISMS OF SPATTER FORMATION

While spatter can result from multiple interacting factors, the fundamental mechanism relates to the balance between the internal force F_N of the molten nugget and the electrode force F_E . Spatter occurs when:

$$F_N \geq F_E \quad (3)$$

Here, F_N is driven by the internal pressure of the molten metal, which increases with greater heat input Q according to Joule's law. Excessive heat input, due to high resistance, current, or time, can therefore lead to spatter. Spatter is undesirable as it poses a safety hazard and contaminates both the equipment and the workpiece. Conversely, insufficient heat may result in undersized nuggets or poor bonding (Senkara et al., 2004, p. 125).

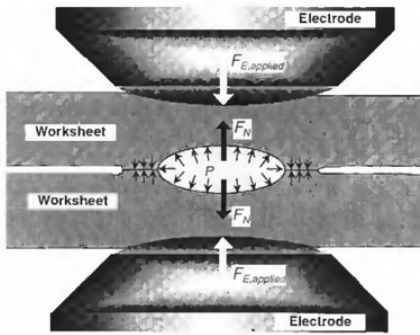


Figure 3. Forces acting during resistance spot welding under ideal conditions (Senkara et al., 2004, p. 125).

2.2.4. EFFECT OF ELECTRODE MISALIGNMENT

Electrode misalignment can further promote spatter. A lateral displacement d of the electrode force from the nugget

center reduces the effective electrode force:

$$F_E = \frac{r-d}{r} \cdot F_{E, \text{applied}} \quad (4)$$

where r is the nugget radius. Reduced effective force means less containment of molten metal, increasing the likelihood of expulsion (Senkara et al., 2004, p. 125).

2.2.5. PULLOUT FAILURE TEST AND FAILURE MODES

The previous sections have outlined the physical principles and process characteristics of resistance spot welding. In the following, we focus on the pullout failure mode of resistance spot welds during tensile-shear testing, as this forms the basis for the physics-informed model developed in our experiments. The objective is to introduce the mechanical background and the analytical relationship required later for the implementation of the physics constraint. This formulation is based on the work of Pouranvari and Marashi (Pouranvari & Marashi, 2021).

One of the most common mechanical tests for evaluating resistance spot weld quality is the tensile-shear test, in which a welded coupon is subjected to a uniaxial tensile load until failure. The goal is often to achieve a *pullout failure* mode, where the weld nugget is pulled out from one of the sheets, as this typically indicates good weld strength and ductility.

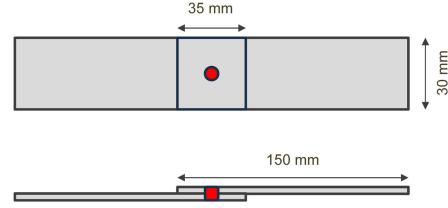


Figure 4. Tensile-shear test sample dimension (Domínguez et al., 2025).

According to Pouranvari et al. (Pouranvari & Marashi, 2021), the primary failure modes observed in tensile-shear testing of spot welds are:

- **Interfacial failure (IF):** Fracture occurs along the faying interface between sheets, often due to insufficient fusion zone diameter or the presence of defects. This mode is generally undesirable, as it indicates lower load-bearing capacity.
- **Partial interfacial-pullout failure (PIF):** A mixed mode where part of the fracture occurs along the interface and part involves nugget pullout.
- **Pullout failure (PF):** The preferred mode, where the weld nugget is completely pulled out from one sheet, leaving a corresponding hole. This mode requires adequate fusion zone size and material ductility.

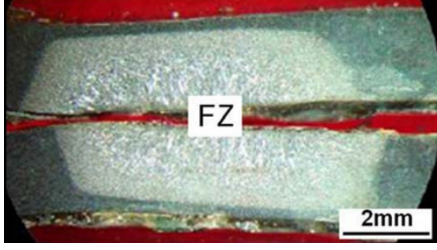


Figure 5. Crack propagation through weld fusion zone in interfacial failure mode (Pouranvari & Marashi, 2021).

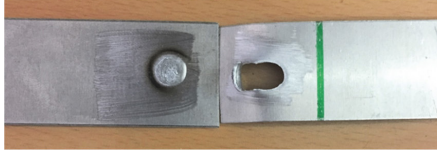


Figure 6. Pullout failure mode of a RSW joint (Ziedan & Doos, 2018).

The nugget diameter is the most important parameter for determining its mechanical behavior. Various industrial standards specify a minimum weld size for a given sheet thickness (Pouranvari & Marashi, 2021). Often is following guideline recommended for low carbon spot weld (Society et al., 1997):

$$d = 4\sqrt{t} \quad (5)$$

- d is the recommended minimum weld diameter in millimeters,
- t is the sheet thickness in millimeters.

For small nuggets, failure can be viewed as a competitive process between crack propagation through the fusion zone (interfacial failure mode) and nugget pullout. The governing failure mode is the one requiring the lower critical load to initiate fracture.

In the tensile-shear testing of resistance spot welds, the driving force for interfacial failure is the shear stress acting along the interface between the two sheets, whereas the driving force for pullout failure is the tensile stress acting around the weld nugget circumference (Pouranvari & Marashi, 2021). Each mode has a characteristic critical load, and fracture occurs in the mode that reaches its critical value first.

For the interfacial failure (IF) mode, the peak load can be estimated by modeling the weld nugget as a cylinder with diameter d and height $2t$. The corresponding failure load F_{IF} is given in the work by Pouranvari et al. (Pouranvari & Marashi, 2021):

$$F_{IF} = \frac{\pi}{4} d^2 \tau_{WN} \quad (6)$$

where:

- d is the weld nugget diameter (mm),
- t is the sheet thickness (mm),
- τ_{WN} is the ultimate shear strength of the weld nugget (MPa), which for steels can be approximated as 0.7-0.8 of the ultimate tensile strength (UTS) (Pouranvari & Marashi, 2021).

For the pullout failure (PF) mode, plastic deformation occurs in the material surrounding the nugget. Large plastic strains cause significant geometric changes, making a closed-form analytical solution that accounts for finite deformation challenging. To derive a practical relationship, it is assumed that pullout failure initiates when the maximum radial tensile stress at the nugget circumference reaches the ultimate tensile strength of the failure location. Under this assumption, the failure load F_{PF} can be expressed as:

$$F_{PF} = \pi (d + 2x) t \sigma_{FL} \quad (7)$$

where:

- x is the distance from the fusion boundary to the pullout failure location (mm),
- σ_{FL} is the ultimate tensile strength at the pullout failure location (MPa).

Finite element simulations by Satoh et al. (Satoh et al., 1991) indicate that the maximum plastic strain occurs at a distance from the fusion boundary on the order of the sheet thickness. As a first approximation, x can be taken as equal to the sheet thickness when the failure occurs in the base metal region.

Since the failure mode is determined by the lower critical load, pullout failure will occur if:

$$F_{PF} < F_{IF} \quad (8)$$

Conversely, if $F_{IF} < F_{PF}$, the weld will fail via interfacial fracture.

2.3. Integrating Physics into Machine Learning Model

The preceding sections outlined the physical principles governing resistance spot welding and their mathematical formulation. This section reviews approaches for incorporating such physical knowledge into machine learning models. The primary objective is to integrate domain knowledge to achieve higher predictive accuracy, improved robustness, and potentially better computational performance, while ensuring consistency with established physical principles.

A prominent concept in this field is *physics-informed machine learning*, which embeds physical laws, such as partial differential equations, conservation laws, or empirical constitutive relationships, directly into the learning process. As described by Karniadakis *et al.* (Karniadakis *et al.*, 2021), physics-informed approaches reduce and structure the hypothesis space of machine learning models by enforcing known physical relationships. These constraints can be implemented as hard conditions, soft penalties in the loss function, or inductive biases in the model architecture, guiding the model toward solutions that are both accurate and physically plausible.

Integration of physics into machine learning models can be framed as introducing biases that guide the learning process toward physically consistent solutions (Karniadakis *et al.*, 2021). Karniadakis *et al.* distinguish three primary forms:

- **Observational biases:** Integrate physical knowledge directly into the training data, for example, through measurements that reflect the underlying physical laws or through data augmentation techniques designed to preserve these laws.
- **Inductive biases:** Embedding prior assumptions about the physical system into the model architecture to ensure that predictions implicitly satisfy certain physical constraints such as symmetries or conservation laws.
- **Learning biases:** Adjusting how the model is trained, for example by adding extra terms to the loss function or applying constraints during optimization, so that the resulting predictions are encouraged to follow the governing physical laws.

These approaches can be combined to form hybrid models. Hybrid approaches combine different forms of physics integration to leverage the strengths of each. One notable strategy, discussed by Karniadakis *et al.* (Karniadakis *et al.*, 2021), is the use of *multi-fidelity models*, where a low-fidelity physics-based model provides a baseline prediction that is subsequently refined by a data-driven component. This can be implemented in various ways, such as generating synthetic low-fidelity data for augmentation, embedding the low-fidelity model within a learning architecture, or learning the *model discrepancy* systematic error between the physics-based prediction and the observed measurements.

In this residual-learning paradigm, the machine learning model does not attempt to predict the target variable from scratch. Instead, it is trained to estimate the bias between the low-fidelity prediction (e.g., from an analytical or empirical formula) and the measured ground truth. The corrected prediction is then obtained by adding the learned bias to the original physics-based estimate. This *gray-box* strategy preserves interpretability and the extrapolative structure of the

physical model while allowing the data-driven component to capture unmodeled effects such as material variability or process noise.

In the context of this work, this approach will be employed to integrate a physically derived formula for the pull-out force $F_{\text{pull,phys}}$ into the learning process. An XGBoost model will be trained to predict the difference between $F_{\text{pull,phys}}$ and the measured pull-test force, thereby forming a hybrid physics-machine learning predictor.

3. Methodology

3.1. Dataset Description

The experiments in this work are based on the *Resistance Spot Welding Insights* dataset (Domínguez *et al.*, 2025), which was generated through a controlled study on a small-scale resistance spot welding machine. A total of 495 welded joints were produced by varying process parameters such as electrode force, electrode angle, welding time, and welding current. The welded coupons were prepared in accordance with the AWS D8.9 standard, ensuring consistent specimen geometry for mechanical testing.

For each weld, process parameters were measured using dedicated instrumentation. A strain gauge recorded electrode force at 10 Hz, and a current transformer measured welding current over the same sampling rate. The welding time was controlled via the machine’s programmable logic, while electrode angle and applied pressure were set according to the experimental plan. In addition, the thickness of each of the two overlapping sheets (*Thickness A* and *Thickness B*) was measured individually, allowing for combinations of equal or different sheet gauges. All sheets were made from AISI 1010 carbon steel.

Following welding, each sample underwent a standardized tensile-shear pull test to determine the *pull-test force* (N) and a measurement of the weld *nugget diameter* (mm). Each weld was also assigned a categorical quality label (*good*, *bad*, or *explode*) based on visual inspection. The full set of recorded variables is:

- **Sample ID** - unique identifier for each weld.
- **Pressure (PSI)** - pneumatic pressure applied to the welding cylinder.
- **Welding Time (ms)** - duration of current application.
- **Electrode Angle (°)** - inclination of electrodes relative to the sheet plane.
- **Electrode Force (N)** - force applied to the sheets during welding.

- **Welding Current (A)** - electrical current through the weld.
- **Thickness A (mm), Thickness B (mm)** - thicknesses of the two sheets; these may differ.
- **Material** - composition of the sheet material (here: AISI 1010 carbon steel).
- **Pull-Test Force (N)** - mechanical resistance measured in tensile-shear testing.
- **Nugget Diameter (mm)** - measured weld nugget size.
- **Category** - categorical weld quality label.
- **Comments** - free-text notes from the experiment.

In its original form, the dataset also contains thermographic and surface images of the welds (front and back views, 990 images in total). These are excluded in the present work, which focuses solely on the tabular process and test data.

In preparation for the physics-informed hybrid model, the dataset is augmented with a calculated pull-out force $F_{\text{pull,phys}}$, derived from the analytical formulation in Section 2.3. This additional feature, along with other preprocessing steps such as data cleaning, feature selection, and dataset splitting, is described in detail in the following section.

3.2. Data Preprocessing and Preparation

The raw dataset contained multiple measurements for the same **Sample ID**, typically corresponding to repeated force readings under identical welding conditions. The measured *pull-test force* was identical for all entries of a given sample. To ensure one-to-one correspondence between samples and rows, only the *last* entry for each **Sample ID** was retained.

In addition, a combined sheet thickness feature was created:

Thickness A+B (mm)

Samples with a combined thickness exceeding the 99th percentile were removed to exclude clear outliers or faulty measurements.

Several variables were excluded from the feature set for the baseline models:

- **Material** - constant across all samples (*AISI 1010 carbon steel*).
- **Comments** - unstructured text without predictive value.
- **Nugget Diameter (mm)** - requires destructive testing, unavailable at inference time.

- **Sample ID** - unique identifier without physical relevance.

For the physics-informed XGBoost residual learning model, an additional preprocessing step was performed: the analytically calculated pull-out force $F_{\text{pull,phys}}$ was computed from the physical formulation in Section 2.3 and stored in a *separate* CSV file. This column was included only in the dataset used for training the residual learning model, where the target variable was defined as:

$$y = F_{\text{pull,test}} - F_{\text{pull,phys}},$$

For all other models (Random Forest, XGBoost, TabPFN), $F_{\text{pull,phys}}$ was not present in the training data.

After preprocessing, the dataset was randomly split into:

- **Training set:** 275 samples.
- **Development set:** 70 samples (for model tuning and selection).
- **Test set:** 149 samples (held out until final evaluation).

The split ratio was 30% test and 70% train+dev, with the latter further divided in an 80/20 proportion.

3.2.1. EXPLORATORY DATA ANALYSIS

Figures 8 and 9 illustrate the distribution of the welding current and welding time, respectively. Both variables exhibit a wide range of values, with several extreme outliers, particularly in the current measurement, which may correspond to faulty settings or atypical welding scenarios. The welding time distribution is comparatively more compact, centred around ≈ 800 ms.

Figure 10 presents the pull-test force as a function of combined sheet thickness, grouped into discrete bins. For thinner sheet combinations ($\approx 1.2\text{--}1.4$ mm), the pull-test forces cluster around 3000 N, with relatively low variability. In contrast, thicker sheet combinations (≥ 1.8 mm) show a much wider spread, with maximum forces exceeding 5500 N, indicating stronger weld joints but also greater variability in weld quality.

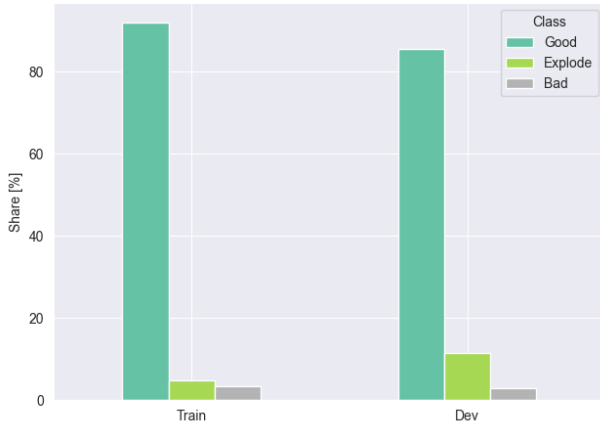


Figure 7. Distribution of quality classes in the training and development sets.

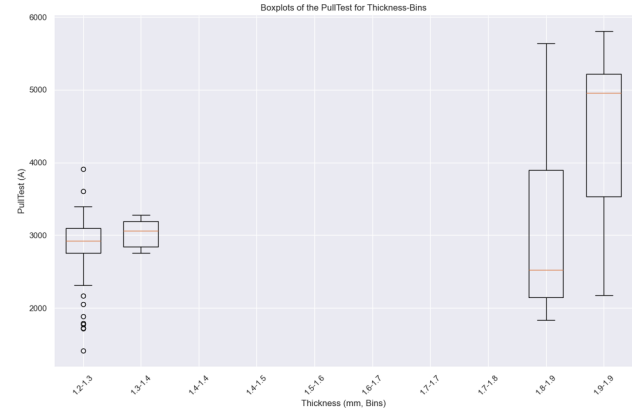


Figure 10. Boxplot of the Pulltest for Thickness-Bins

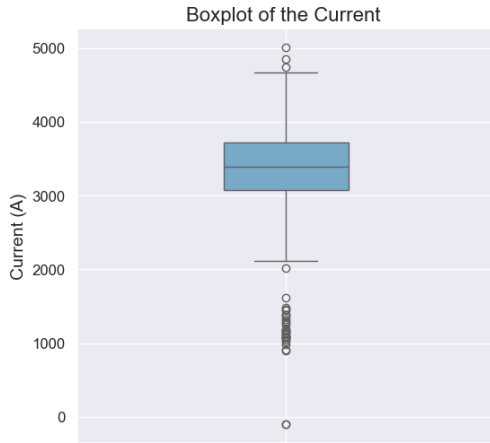


Figure 8. Boxplot of welding current values.

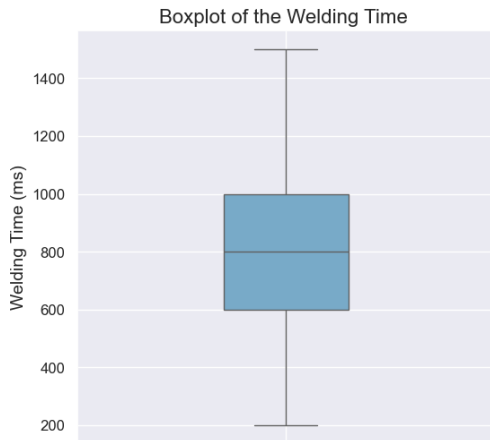


Figure 9. Boxplot of welding time values.

3.3. Models

3.3.1. XGBOOST BASELINE AND HYPERPARAMETER OPTIMIZATION

As one of the baseline models, an `XGBOOST` regressor was implemented, given its proven ability to handle non-linear relationships and heterogeneous feature sets in tabular manufacturing data. The initial baseline configuration used standard parameters without any task-specific tuning.

To improve predictive performance, a hyperparameter optimization was carried out using the `Hyperopt` library with the Tree-structured Parzen Estimator (TPE) algorithm. The optimization objective was the minimization of the root mean squared error (RMSE) on the training set. The search space included:

- **max depth** - maximum depth of individual trees.
- **learning rate** - shrinkage factor controlling the contribution of each tree.
- **subsample** - fraction of the training data used per boosting iteration.
- **gamma** - minimum loss reduction required to make a further partition.
- **reg alpha** - L1 regularization term on weights.
- **reg lambda** - L2 regularization term on weights.
- **colsample bytree** - fraction of features sampled per tree.
- **min child weight** - minimum sum of instance weight needed in a child node.
- **n estimators** - number of boosting rounds.

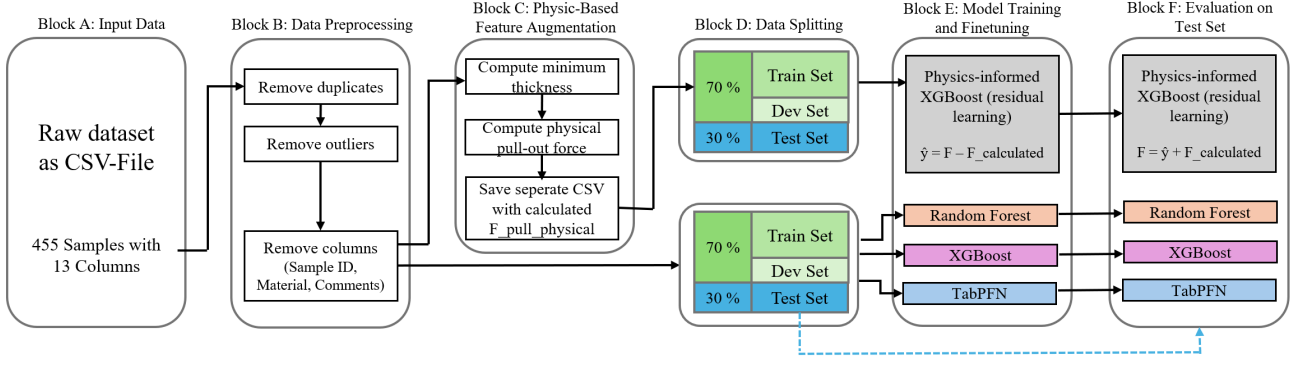


Figure 11. Overall workflow of the proposed method, including preprocessing, separate creation of the physics-augmented dataset, model training, and evaluation.

To evaluate the impact of tuning, the baseline model and six tuned variants (`tuned_v1`–`tuned_v6`) were compared via 5-fold cross-validation on the combined training and development datasets. Figure 12 shows the RMSE (mean $\pm$ standard deviation) for each variant.

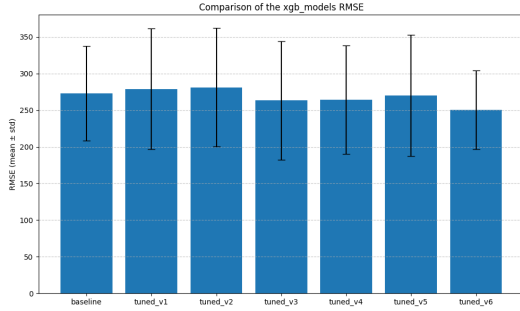


Figure 12. Comparison of baseline and tuned XGBoost models using 5-fold cross-validation. Error bars indicate standard deviation.

The best-performing configuration (`tuned_v6`) achieved an average RMSE of 250.29 ± 57.19 N, representing a notable improvement over the baseline (273.17 ± 64.54 N). The parameter for this configuration was:

- **n estimators** = 280
- **max depth** = 2
- **learning rate** = 0.0549
- **subsample** = 0.8215
- **gamma** = 8.7214
- **colsample bytree** = 0.9141
- **min child weight** = 5

- **reg alpha** = 10
- **reg lambda** = 0.5425
- **random state** = 42

This reduction in error demonstrates the effectiveness of hyperparameter tuning for adapting the XGBoost model to the specific characteristics of the resistance spot welding dataset.

To gain further insight into the predictive behaviour of the tuned XGBoost model, the feature importances were extracted using the built-in *gain*-based importance metric. This metric quantifies the average improvement in the model’s objective function brought by splits on a given feature, aggregated over all trees in the ensemble.

Figure 13 shows the resulting importances, normalized to sum to 1. The *Welding Time* emerged as the dominant predictor by a large margin, followed by *Angle* and *Pressure*, indicating that temporal and geometric welding parameters have the most influence on predicting the pull-test force. Mechanical parameters such as *Force* and material thicknesses (*Thickness A*, *Thickness B*, and their sum) play a secondary, but still notable, role. The relatively lower importance of the *Current* variable suggests that, within the observed parameter ranges, variations in electrical current were less critical for the model’s performance than temporal and geometric factors.

These importance rankings align partially with domain expectations: welding time is known to strongly affect heat input and nugget formation, while electrode angle and applied pressure influence contact resistance and nugget geometry. However, the lower ranking of welding current contrasts with common process understanding, possibly reflecting the limited variation of current within the available dataset.

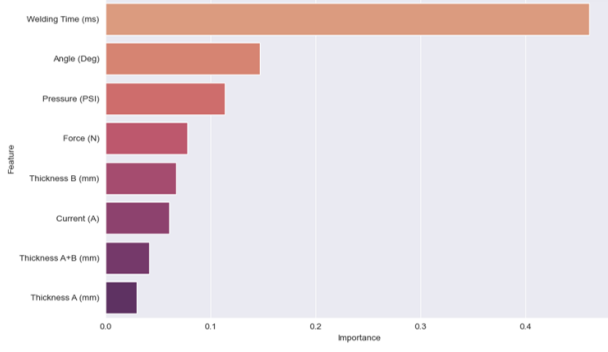


Figure 13. Feature importance of the tuned XGBoost model, measured by average gain across all splits.

3.3.2. RANDOM FOREST BASELINE AND HYPERPARAMETER OPTIMIZATION

As a second non-parametric baseline, we trained a *Random Forest* regressor. Random forests are well suited for heterogeneous tabular data and provide strong variance reduction via bagging.

Tuning protocol. Hyperparameters were optimized with Hyperopt (TPE), minimizing RMSE computed via 5-fold cross-validation on the *training set*. The development set was kept strictly separate and used only after tuning for model checking. The search covered:

- **n estimators** - number of trees (100-300).
- **max depth** - maximum tree depth (5-30).
- **min samples split** - minimum samples to split an internal node (2-10).
- **min samples leaf** - minimum samples at a leaf (1-10).
- **max features** - features considered at each split (None, sqrt, log2).
- **bootstrap** - sampling with/without replacement (True/False).

Best configuration. The best RF configuration (tuned_v3) selected by CV was:

- **n estimators** = 100
- **max depth** = 23
- **min samples split** = 10
- **min samples leaf** = 3
- **max features** = None

- **bootstrap** = True

- **random state** = 42

Cross-validated performance. The baseline configuration and three tuned variants (tuned_v1-tuned_v3) were compared using 5-fold cross-validation on the combined training and development dataset. Figure 14 reports RMSE values (mean $\pm$ std). The tuned_v3 configuration achieved the lowest mean RMSE of 251.20 ± 56.95 N, improving over the baseline RMSE of 266.30 ± 76.86 N. The selected hyperparameters for tuned_v3 included relatively deep trees regularized through larger leaf and split size thresholds, the use of all features per split (max_features=None), and bootstrap aggregation.

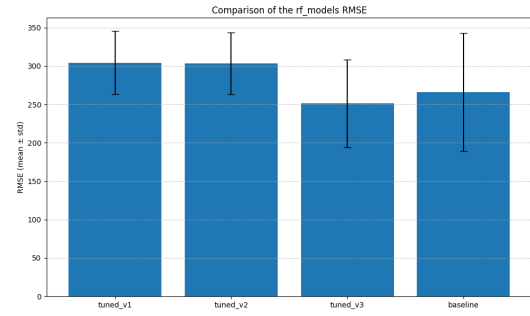


Figure 14. Random Forest model comparison via 5-fold cross-validation (RMSE mean $\pm$ std).

TABPFN

TabPFN was included as a foundation-model baseline for tabular regression. The model was used in its standard configuration without task-specific hyperparameter tuning. Training was performed on the training split using the same preprocessed feature set as for the tree-based baselines. The implementation employed the TabPFNRegressor. On the development set, TabPFN achieved an RMSE of 238.6 N, which is competitive with the tuned tree-based models and outperforms the tuned Random Forest.

3.4. Physic-informed XGBoost implementation

To incorporate prior physical knowledge into the learning process, a physically motivated estimate of the pull-out force $F_{\text{pull,phys}}$ was computed for the *interfacial failure* (IF) mode. The calculation follows the cylindrical nugget model described in Section 2.3, where the nugget diameter d is required. Although the dataset contains measured nugget diameters, these are obtained through destructive testing after the pull-out test and are therefore unavailable at inference time. Consequently, d was approximated from the minimum sheet thickness t_{min} of the weld stack using

$$d \approx 4\sqrt{t_{\text{min}}},$$

an empirical relationship motivated by resistance spot welding literature.

The IF failure load was then estimated via

$$F_{\text{IF}} = \frac{\pi}{4} d^2 \tau_{\text{WN}},$$

where τ_{WN} is the ultimate shear strength of the weld nugget. For AISI 1010 steel, τ_{WN} was set to 292 MPa, corresponding to 0.8 of the ultimate tensile strength (365 MPa) (Pouranvari & Marashi, 2021; Components, n.d.).

Figure 15 compares the ground truth pull-out force with two physical model estimates: one based on the measured nugget diameter and one using the thickness-based approximation. As expected, the model using the measured diameter achieves lower error (RMSE = 524.2 N) compared to the approximation (RMSE = 708.3 N), reflecting the information loss when substituting the diameter with a thickness-based estimate.

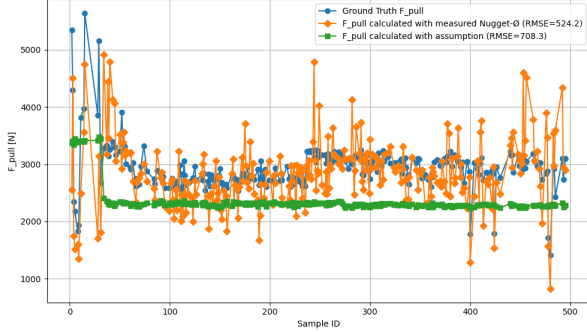


Figure 15. Comparison of ground truth F_{pull} with physically calculated values using measured and approximated nugget diameters.

Visual inspection of the analytic interfacial-failure prediction F_{IF} revealed a systematic deviation from the measured pull-test force F_{pull} (cf. Fig. 15). While the analytic formulation captures the main mechanical trend, it cannot account for complex process effects such as electrode wear, surface contamination, or subtle geometric variations. Directly modeling F_{pull} from scratch would therefore require the regression model to learn both the underlying physical law *and* the data-specific deviations, which can be sample inefficient in a small data regime.

To address this, a residual learning strategy was adopted. Instead of predicting the absolute force, the model learns the residual

$$r := F_{\text{pull}} - F_{\text{IF}},$$

where F_{IF} is the low-variance physical approximation from Section 2.3 under the IF-mode assumption. At inference time, the final prediction is reconstructed as

$$\hat{F}_{\text{pull}} = F_{\text{IF}} + \hat{r}.$$

This decomposition allows the physical term to encode the dominant structural dependence on sheet thickness, while the machine learning model focuses solely on systematic deviations caused by process-specific factors.

For the residual model, all features unavailable at inference were excluded, and F_{IF} itself was not part of the input feature set. The target variable was defined as r , the residual between the measured and the analytic force.

Hyperparameter tuning and evaluation. Two residual-learning XGBoost models were compared: a baseline configuration and a tuned variant optimised via Hyperopt over a predefined search space for key hyperparameters (tree depth, learning rate, regularisation terms, column and row subsampling, and number of estimators). Performance was assessed using 5-fold cross-validation on the combined training+development set. Importantly, RMSE was computed on the reconstructed force $\hat{F}_{\text{pull}}$, not on the residual itself.

The tuned variant reduced the average RMSE compared to the baseline in cross validation:

- Baseline: 253.32 ± 71.43 N
- Tuned_v1: 231.88 ± 69.56 N

3.4.1. COMPARISON OF THE MODELS

To directly compare the predictive performance of all considered approaches, Figure 16 shows the cumulative distribution functions (CDFs) of the absolute errors on the development set. For any error threshold on the horizontal axis, the CDF value indicates the proportion of predictions with absolute error less than or equal to that threshold, providing a more informative view of the error distribution than a single summary statistic. All results in this subsection refer exclusively to the development set and not to the held-out test set.

The physics-guided residual XGBoost achieves the best overall performance, exhibiting the steepest CDF and the thinnest upper tail, i.e., a high share of low-error predictions and markedly fewer large deviations (**RMSE:** 231.9 N). TabPFN also performs well, showing competitive performance close to the physics-guided model (**RMSE:** 238.6 N). The tuned plain XGBoost follows closely (**RMSE:** 245.1 N), and the tuned Random Forest achieves (**RMSE:** 256.4 N). For TabPFN, plain XGBoost, and Random Forest, occasional individual predictions with high absolute error remain visible, whereas the physics-guided XGBoost almost entirely avoids such large deviations.

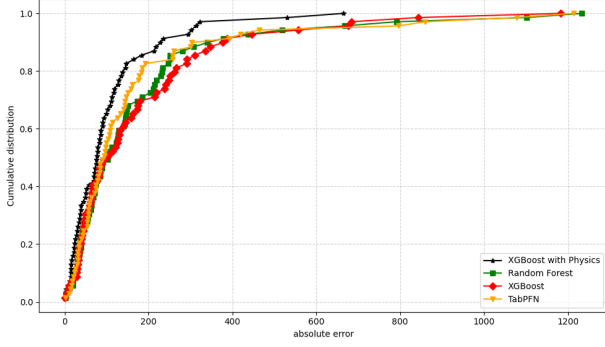


Figure 16. CDFs of absolute errors for all evaluated models on the development set.

4. Results

4.1. Performance Comparison across models

After finalizing the model configurations based on the development set, all approaches were evaluated on the test set. The performance was quantified using the root mean squared error (RMSE) in N, with lower values indicating better predictive accuracy. Table 1 summarizes the results, and Fig. 17 shows the cumulative distribution functions (CDF) of absolute errors.

Table 1. RMSE on the held-out test set for all compared models.

Model	RMSE [N]
Random Forest	226.98
XGBoost	240.40
TabPFN	227.81
XGBoost with Physics	226.75

The physics-guided residual XGBoost achieves the lowest RMSE (226.75 N), closely followed by the Random Forest (226.98 N) and TabPFN (227.81 N). The plain XGBoost model yields a slightly higher RMSE (240.40 N). While the performance differences are small in absolute terms, the CDF curves in Fig. 17 reveal that the physics-guided approach produces fewer large-error predictions, consistent with the development set results. TabPFN also performs competitively, matching the physics-guided model across most of the error range.

4.2. Error Analysis and Interpretation

While the RMSE differences between models on the test set are small in absolute terms (Table 1), the CDF in Fig. 17 reveals that the physics-guided residual XGBoost yields a slightly steeper ascent in the high-accuracy region, indicating fewer large-error predictions. Random Forest unexpectedly matches its RMSE performance on the test set, despite

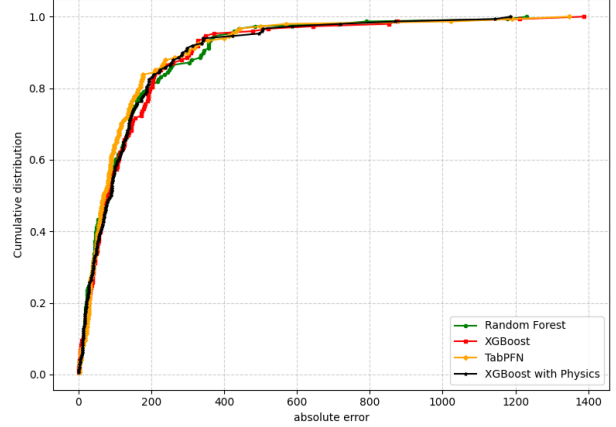


Figure 17. Cumulative distribution of absolute errors on the test set for all models.

performing worse on the development set, suggesting that its ensemble of unpruned trees may generalize well under certain process parameter distributions. TabPFN also performs competitively, confirming its suitability for small tabular manufacturing datasets.

Relation to Feature Importance. The tuned XGBoost feature importance analysis (Section 3.3.1) showed *Welding Time*, *Angle*, and *Pressure* as the dominant predictors, followed by *Force* and material thicknesses. These rankings align with welding domain knowledge: temporal and geometric parameters strongly influence nugget formation and pull-test force, while current variation played a smaller role within the observed parameter range. The competitive performance of the physics-guided model suggests that augmenting the feature set with the physically computed $F_{\text{pull,phys}}$ effectively captures relevant nugget strength characteristics.

Analysis of High-Error Points. The largest absolute errors are concentrated in a small set of test samples (Table 2), with errors exceeding 450 N and up to 1241 N. A majority of these cases are labelled as *Explode*, a category that is underrepresented in the training data (13 samples in the training set, 8 in the development set). This imbalance limits the model’s ability to learn the relationship between extreme welding conditions and pull-test force for such cases.

From a physical perspective, *Explode* events correspond to excessive heat input Q in the nugget, which increases the internal force F_N of the molten metal until it exceeds the electrode force F_E :

$$F_N \geq F_E. \quad (9)$$

As described in Section 2.2.3, this results in expulsion of molten metal (spatter), reducing the effective nugget volume and thereby the measured pull-test force. Because the

nugget geometry is altered in a non-systematic way during such events, the prediction task becomes considerably harder, particularly in data regimes where such cases are rare.

Table 2. Highest absolute prediction errors in the test set.

Category	Time [ms]	Angle [°]	Ground Truth [N]	Error [N]
Explode	600	15	1726	1241
Explode	600	15	2168	840
Good	1500	0	3898	530
Explode	1500	15	4952	483
Good	1500	15	5278	452

Overall, this error analysis highlights that while the models perform similarly on the majority of the test set, performance gaps are more pronounced in rare, extreme conditions, particularly for the *Explode* category, where physical anomalies in the weld nugget geometry reduce the predictive power of purely data driven models.

5. Discussion

The comparative evaluation on the held-out test set revealed that all approaches achieved similar RMSE values in the range of 226.75–240.40 N, with the physics-guided residual XGBoost achieving the lowest error, closely followed by Random Forest and TabPFN (Table 1). While the absolute performance differences are small, the CDF curves (Fig. 17) indicate that the physics-guided model produces fewer large-error predictions, consistent with the development set trends.

5.1. Performance of TabPFN

One notable observation is the strong performance of TabPFN, which achieved an RMSE of 227.81 N without any hyperparameter tuning or task specific adaptations. This is particularly striking given the relatively small size of the dataset and the heterogeneous nature of the features, conditions under which TabPFN’s transformer based architecture and pretraining are expected to excel. It is reasonable to speculate that, if additional data modalities (e.g., categorical metadata or text-based process notes) were available, TabPFN could leverage them more effectively than tree-based models due to its ability to process mixed-type features natively. However, such multimodal integration was not explored in the present work.

5.2. Trade-offs of the Physics-Guided Approach

The physics-guided residual XGBoost incorporated an estimate of the interfacial failure pull-out force $F_{\text{pull,phys}}$ as described in Section 2.3. While this model achieved the lowest RMSE, it came with higher computational effort: additional feature engineering, derivation of the physical prior, and model retraining for residual learning. Moreover, the

underlying formula carries assumptions and simplifications:

- The nugget is modeled as a perfect cylinder, neglecting irregularities in shape or material distribution.
- The shear strength τ_{WN} is taken as a fixed fraction of the ultimate tensile strength, ignoring possible variations due to local microstructure or thermal history.
- The nugget diameter d is not directly measurable at inference time and was therefore approximated from the minimum sheet thickness t_{min} , introducing additional error.

The impact of these assumptions is evident when comparing the purely physical predictions to ground truth (Fig. 15): using the measured nugget diameter yields an RMSE of 524.2 N, while the thickness-based approximation increases the RMSE to 708.3 N. Even the best physical-only estimate performs considerably worse than the purely data-driven models, highlighting the limitations of the simplified geometric and material model when applied in isolation.

5.3. High-Error Cases and Data Imbalance

The largest prediction errors were concentrated in rare extreme cases, particularly in the *Explode* category. As discussed in Section 2.2.3, these events are caused by excessive heat input, leading to molten metal expulsion and altered nugget geometry. Because such cases are underrepresented in the training data (13 samples in the training set, 8 in the development set), all models struggled to predict their pull-test forces accurately. The physics-guided approach offers some mitigation by encoding domain constraints, but remains limited by the lack of direct nugget geometry measurements during inference.

5.4. Model Selection and Industrial Relevance

From an industrial deployment perspective, the choice between TabPFN and physics-guided XGBoost involves a trade-off:

- TabPFN offers strong out-of-the-box performance with minimal tuning, low development overhead, and potential for multimodal integration.
- The physics-guided approach requires more engineering effort and computational time, but yields a slight performance edge and is more interpretable in terms of known process physics.

In manufacturing settings where explainability and consistency in rare-event scenarios are critical, physics-guided models may be preferable. However, in many industrial processes, deriving accurate analytical formulas is challenging

due to complex interactions, unmeasured variables, and variability in materials and equipment. In such cases, the effort required to develop and validate a reliable physics model may outweigh its benefits. In contrast, for rapid prototyping or situations with diverse feature types, foundation models like TabPFN present a compelling alternative, offering strong performance with minimal manual engineering.

6. Conclusion and Future Work

6.1. Summary of key findings

This work evaluated the foundation model TabPFN in the context of industrial manufacturing, using resistance spot welding (RSW) pull-test force prediction as a representative case study. Four models were compared on the held-out test set: tuned Random Forest, tuned XGBoost, TabPFN, and a physics-guided residual XGBoost integrating an analytical pull-out force estimate. All models achieved similar RMSE values between 226.75 and 240.40 N, with the physics-guided residual XGBoost achieving the lowest RMSE (226.75 N), closely followed by Random Forest (226.98 N) and TabPFN (227.81 N).

TabPFN delivered competitive performance without any task-specific tuning or feature engineering, demonstrating its strong generalization capabilities in small, heterogeneous tabular datasets. The physics-guided model showed a slight performance edge and fewer large-error predictions, particularly in rare-event scenarios, but required significant domain-specific engineering and relied on simplifying assumptions in the analytical formulation. Purely physical predictions were markedly less accurate, highlighting the benefit of combining physics priors with data-driven corrections.

6.2. Practical relevance for manufacturing analytics

From a deployment perspective, the model choice depends on operational priorities:

- For rapid prototyping, heterogeneous data sources, or situations where domain formulas are unavailable, foundation models like TabPFN provide strong baseline accuracy with minimal manual effort and the potential for multimodal integration (e.g., categorical metadata, text notes).
- Where explainability, adherence to physical constraints, and robustness in rare event scenarios are critical, physics guided models remain advantageous despite their higher development cost.

In many industrial processes, deriving accurate analytical models is challenging due to unmeasured variables, material variability, and complex interactions. In such settings, the

additional effort required to design and validate physics based features may outweigh the marginal accuracy gains.

6.3. Potential extensions and research directions

Future work can extend this study along several directions:

- **Hybrid foundation-physics models:** Integrate physics residual learning into TabPFN, combining its in-context learning capabilities with domain priors.
- **Generalization studies:** Evaluate the models on a broader range of manufacturing processes and datasets to assess robustness beyond resistance spot welding.
- **Richer categorical data integration:** Apply TabPFN to datasets with a higher proportion of categorical features or mixed data types to explore its strengths in handling heterogeneous inputs.
- **Beyond regression:** Investigate TabPFN's performance in classification tasks relevant to manufacturing quality control, such as weld defect detection.

These directions offer promising opportunities to further assess both the predictive performance and the domain adaptability of AI models in manufacturing analytics.

Code Availability

The implementation and all experiment scripts are available at <https://github.com/SamuelRusch/tabPFN-industrial-eval>.

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